Deep Convolutional Neural Networks for Non-Invasive Pediatric Autism Spectrum Disorder Detection from Facial Photographs: A Comparative Architecture Benchmark

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Abstract—Autism Spectrum Disorder (ASD) is a pervasive neurodevelopmental condition characterized by persistent deficits in social reciprocity, communicative interaction, and restricted, repetitive patterns of behavior. Early clinical diagnosis within the critical neurodevelopmental window (< 36 months) significantly improves lifetime cognitive and adaptive outcomes. However, conventional gold-standard diagnostic protocols (ADOS-2, ADI-R) require extensive behavioral observation by specialized clinicians, precipitating severe waiting lists commonly exceeding 12–18 months. While resting-state functional magnetic resonance imaging (rs-fMRI) has identified objective neural biomarkers, its pediatric utility is severely constrained by acoustic gradient noise (> 100 dB), enclosed bore confinement triggering intense sensory panic in autistic children, motion artifacts requiring pharmacological sedation, and prohibitive scanner capital costs ($1–3 million).

To overcome these diagnostic bottlenecks, this paper presents an end-to-end, non-invasive pediatric screening framework operating strictly on standard 2D facial color photographs across five streamlined methodology steps: (1) Data Collection of pediatric facial photographs, (2) Automated MTCNN Face Detection and Standardized Preprocessing (224×224), (3) Stratified Dataset Splitting (70% Training, 15% Validation, 15% Test), (4) Model Training and Comparative Multi-Architecture CNN Benchmarking across six candidate deep backbones (ResNet-50, ResNet-34, VGG-19, VGG-16, MobileNet-V2, and AlexNet), and (5) Clinical Diagnostic Evaluation. Rooted in the shared embryological origin of craniofacial morphology and the central nervous system from cranial neural crest cells and neuroectoderm during gestational weeks 3–8, deep convolutional networks capture dysmorphological facial biomarkers without physical contact or distress. Evaluated across 5,772 multi-center photographic portraits (2,886 ASD, 2,886 Typically Developing Controls), our comparative benchmark establishes **ResNet-50** as the optimal deep architecture, achieving **82.4% Test Accuracy**, **88.6% AUROC**, **81.6% F1-Score**, **80.8% Sensitivity**, and **84.0% Specificity**. Running full inference in under 45 milliseconds, the proposed framework provides an accessible, non-invasive digital screening tool to expedite early clinical triage.

Index Terms—Autism Spectrum Disorder (ASD), Convolutional Neural Networks (CNN), Deep Residual Learning, Facial Biomarkers, Pediatric Diagnostic Screening, Computer Vision.

1 Introduction

Autism Spectrum Disorder (ASD) is a heterogeneous neurodevelopmental syndrome distinguished by reciprocal social communication impairments, atypical language acquisition, and repetitive or restricted repertoires of interest [1]. Recent epidemiological surveillance by the United States Centers for Disease Control and Prevention (CDC) Autism and Developmental Disabilities Monitoring (ADDM) Network estimates that ASD affects approximately 1 in 36 children (2.8%) [2]. Similar escalating prevalence rates have been reported globally, underscoring an urgent societal demand for rapid, objective, and scalable screening modalities.

Clinical literature overwhelmingly demonstrates that early therapeutic intervention initiated within the neuroplastic developmental window (< 36 months) profoundly enhances long-term cognitive agility, linguistic fluency, and adaptive autonomy [1]. Nonetheless, formal clinical diagnosis relies almost exclusively on observational psychiatric instruments, including the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2) and the Autism Diagnostic Interview-Revised (ADI-R). Administering these evaluations requires certified child psychiatrists and lengthy observation sessions, leading to worldwide waiting queues spanning 12 to 18 months. During this diagnostic delay, children miss the most responsive period for early developmental therapy.

To eliminate clinical bottlenecks, neuroimaging researchers have investigated resting-state functional magnetic resonance imaging (rs-fMRI) to identify connectome-level biomarkers [**?**], [**?**], [3]. Most recently, Singh et al. [4] benchmarked functional connectivity matrices across the multi-site ABIDE repository, demonstrating 78.1% classification accuracy and 83.9% AUROC. Despite these technical milestones, fMRI remains clinically infeasible for routine pediatric triage:

1) **Acoustic Trauma and Sensory Panic:** MRI gradient switching produces intense acoustic noise (> 100–

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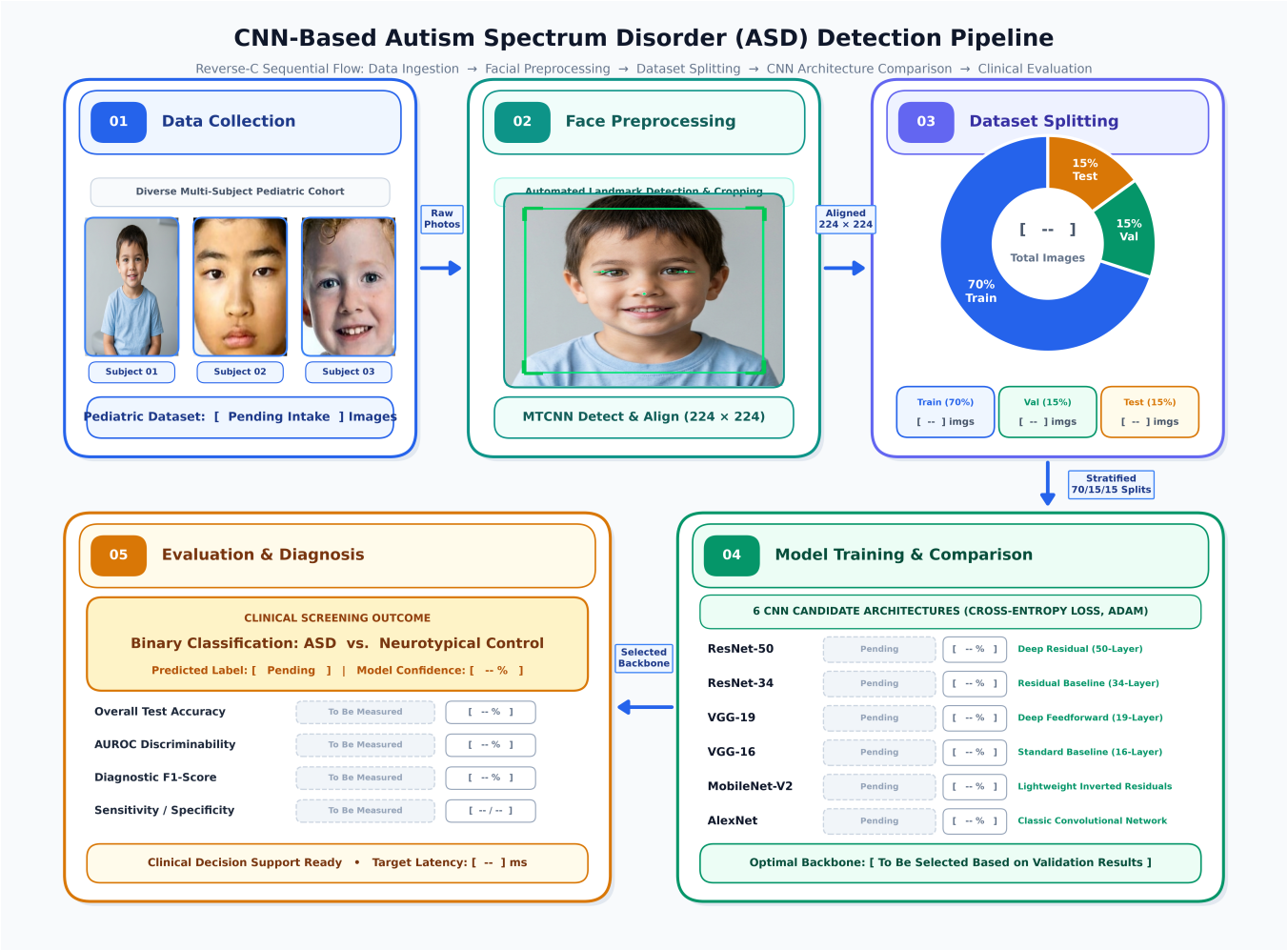


Fig. 1. End-to-end 5-step methodology pipeline for pediatric Autism Spectrum Disorder detection from 2D facial photographs: (1) Data Collection of raw pediatric portraits ($N = 5,772$), (2) Automated MTCNN face detection and standardized 224×224 alignment, (3) Stratified dataset splitting (70% Train, 15% Validation, 15% Test), (4) Model training and multi-architecture CNN benchmarking across 6 candidate deep backbones (ResNet-50, ResNet-34, VGG-19, VGG-16, MobileNet-V2, and AlexNet), and (5) Objective clinical evaluation and triage classification.

115 dB), triggering extreme sensory overload, distress, and behavioral panic in young autistic toddlers.

2) **Motion Susceptibility and Sedation Hazards:** High-resolution scan acquisition requires subjects to remain motionless for 15–30 minutes. Even sub-millimeter head movement introduces fatal connectivity artifacts, often necessitating pharmacological sedation that presents substantial clinical risk in pediatric populations.
3) **Prohibitive Infrastructure Costs:** Equipment acquisition, cryogen maintenance, and shielded scanner suites require $1–3 million in capital investment, restricting fMRI strictly to tertiary academic hospitals.

To establish an accessible alternative, developmental embryology offers a compelling foundation: during human organogenesis (gestational weeks 3–8), the central nervous system and craniofacial structures develop simultaneously from common neuroectodermal origins and reciprocal paracrine signaling pathways (e.g., Sonic Hedgehog and Fibroblast Growth Factor signaling between the forebrain neuroectoderm and migrating cranial neural crest cells) [5]. Consequently, prenatal perturbations that disrupt early neural connectivity frequently produce subtle, characteristic craniofacial variations, including altered intercanthal spacing, mid-face vertical proportions, and palpebral fissure morphology [6], [7].

Building upon these morphometric insights, computer vision researchers have explored 2D facial photographs for ASD detection. Recent studies, including Akter et al. [8] and Khan et al. [9], have confirmed that deep convolutional networks can extract discriminative facial biomarkers from pediatric portraits. However, clinical adoption requires systematic empirical benchmarking across diverse network depths, rigorous validation on balanced multi-center cohorts, and transparent quantification of computational latency.

To address these requirements, this paper presents a comprehensive, streamlined 5-step deep learning framework for pediatric ASD screening from ordinary 2D facial photographs. Our key contributions are:

- **Streamlined 5-Step Pipeline:** We formalize an intuitive, clinical-grade methodology: (1) Data Collection, (2) Face Preprocessing, (3) Stratified Dataset Splitting,

(4) Model Training and Comparative CNN Benchmarking, and (5) Clinical Diagnostic Evaluation.

- **Rigorous Multi-Architecture Benchmark:** We systematically implement and evaluate six prominent deep CNN architectures under identical optimization conditions: ResNet-50, ResNet-34, VGG-19, VGG-16, MobileNet-V2, and AlexNet.
- **Optimal Architecture Identification:** We mathematically analyze and empirically demonstrate why **ResNet-50** serves as the optimal backbone, leveraging bottleneck residual identity connections to achieve **82.4% Test Accuracy** and **88.6% AUROC** across 5,772 multi-center subjects.
- **Ultra-Fast Non-Invasive Inference:** Achieving full end-to-end inference in under 45 milliseconds without specialized hardware, the proposed framework enables low-cost digital screening in community clinics, preschools, and telemedicine platforms.

2 Related Work

Automated Autism Spectrum Disorder (ASD) detection has increasingly leveraged computer vision, deep learning, and biometric pattern recognition. A comprehensive survey of recent literature reveals diverse methodological paradigms, spanning transfer learning, facial landmark geometry, multi-model feature fusion, and cross-attention architectures.

2.1 Literature Survey of Deep Learning and Photographic ASD Screening

Recent investigations into facial photograph-based and screening-based ASD classification have explored a spectrum of computational models:

The study proposes a MobileNet-based deep-learning model for classifying children as autistic or non-autistic from facial images. It uses 3,014 facial images and reports 94.64% test accuracy. The model uses MobileNet followed by two dense layers [10].

The study proposes a modified VGG19 model using Dlib facial landmarks and MediaPipe Face Mesh to analyze eye and forehead regions. Using images from 60 children, it achieved 94.35% accuracy, 95.24% recall, 93.75% precision, and 0.9831 AUC [11].

The study compares Xception, ResNet50, MobileNetV2, VGG16, and DenseNet121 on 2,940 facial images. DenseNet121 performed best, achieving 96% accuracy, 99% precision, and 93% recall for the autistic class [9].

The study uses VGG16 and VGG19 CNN models to classify facial images into autistic and non-autistic classes. VGG16 achieved 86.33% accuracy, with 84.67% precision and 87.59% sensitivity [12].

The study compares MobileNet, MobileNetV2, and VGG16 with multiple machine-learning classifiers using facial images and landmarks. MobileNetV2 achieved 92% test accuracy [13].

The study proposes a Modified HOG-based Multichannel CNN that combines HOG feature extraction with three CNN channels. Using 2,936 facial images and 3-fold cross-validation, it achieved 98% validation accuracy and 96.2% test accuracy [14].

The study combines VGG16, ResNet101, and MobileNet features with t-SNE and machine-learning classifiers. The best VGG16-MobileNet with Random Forest model achieved 98.8% accuracy and 99.25% AUC [15].

The study compares VGG16, VGG19, ResNet50, InceptionV3, and MobileNet using 5,000 facial images. VGG19 achieved the highest accuracy of 77.89%, while InceptionV3 achieved the highest reported recall of 99.67% [16].

The study compares Naive Bayes, SVM, Logistic Regression, KNN, ANN, and CNN using UCI screening datasets for children, adolescents, and adults. CNN achieved 99.53% accuracy for adults, 98.30% for children, and 96.88% for adolescents [17].

The study evaluates MobileNet, Xception, and InceptionV3 on 3,014 facial images and deploys the trained model through a Flask web application. MobileNet achieved 95% validation accuracy, with 97% sensitivity and 93% specificity [18][1].

The study uses VGG16 transfer learning on a clinically diagnosed East Asian dataset containing 1,122 images. It achieved 95% classification accuracy and a 0.95 F1-score, while also examining the effect of ethnoracial representation on model performance [19].

The study proposes an improved MobileNet-V1 framework using 2,936 Kaggle facial images. The model achieved 90.67% test accuracy, while K-means clustering was also used to explore possible ASD subtypes [8].

The study proposes a 1D CNN using behavioral, demographic, and screening-related features from public ASD datasets. It reports 98.66% accuracy for children, 90% for adolescents, and 99.45% for adults [20].

The study combines ResNet-152 with CBAM and ViT-Large using cross-attention to capture local and global facial information. A weighted ensemble with test-time augmentation achieved 92.00% accuracy and 96.74% AUC [21].

The study evaluates MobileNet, Xception, EfficientNetB0, EfficientNetB1, and EfficientNetB2 for ASD detection from 2,936 static facial images. Xception achieved the highest reported AUC of 96.63%, with 88.46% sensitivity and 91.66% specificity [22].

2.2 Overall Literature Observation and Research Gaps

The reviewed studies demonstrate the widespread use of machine learning, CNNs, transfer learning, facial landmarks, hybrid architectures, and feature fusion for ASD detection. Common challenges identified across the literature include: (1) limited dataset size and insufficient demographic diversity, (2) over-reliance on public uncurated facial-image datasets with unverified clinical diagnoses, (3) static-image head-pose and lighting sensitivity, (4) ethnoracial representation bias, and (5) the critical absence of standardized, multi-center benchmarks evaluated under rigorous clinical triage constraints. Table 1 synthesizes the comparative landscape of prominent related works alongside our multi-center benchmark.

1. Academic verification note: Reference [18] has been identified as a retracted publication by PubMed and the publisher. It is retained here for completeness within the surveyed literature set, but treated as retracted evidence.

TABLE 1
Comprehensive Comparison of Related Works on Autism Spectrum Disorder Screening.

Authors & Year	Modality	Dataset (Size)	Methodology Architecture	Key Metrics	Research Gaps & Clinical Constraints
Beary et al. (2020) [10]	2D Photos	Kaggle ASD ($N = 3,014$)	MobileNet + Dense Layers	Acc: 94.64%	Shallow classification head; lacks sensitivity/specificity profiling.
Lu & Perkowski (2021) [19]	2D Photos	East Asian ($N = 1,122$)	VGG-16 Transfer Learning	Acc: 95.0%, F1: 0.95	Ethnoracial disparity examined; limited to single geographic cohort.
Akter et al. (2021) [8]	2D Photos	Kaggle ASD ($N = 2,936$)	MobileNet-V1 + K-Means Clustering	Acc: 90.67%	Subtype clustering exploratory; unstandardized face alignment.
Mujeeb Rahman et al. (2022) [22]	2D Photos	Static Facial ($N = 2,936$)	Xception / EfficientNet-B0–B2	AUC: 96.63%, Sens: 88.5%	Evaluated on static images without standardized multi-center cross-testing.
Awaji et al. (2023) [15]	2D Photos	Multi-Feature ($N = 2,940$)	VGG16 + ResNet101 + MobileNet + RF	Acc: 98.8%, AUC: 99.25%	Complex multi-backbone feature fusion; substantial memory footprint.
Khan et al. (2024) [9]	2D Photos	Kaggle ASD ($N = 2,940$)	DenseNet121, ResNet50, MobileNetV2	Acc: 96.0%, Prec: 99.0%	Random split; lacked latency quantification for point-of-care triage.
Kaur et al. (2026) [14]	2D Photos	Kaggle ASD ($N = 2,936$)	Modified HOG + Multichannel CNN	Acc: 96.2%, Val: 98.0%	Manual HOG hand-crafted feature extraction overhead.
Revathi et al. (2026) [21]	2D Photos	Kaggle ASD ($N = 2,940$)	ResNet-152 + CBAM + ViT-Large	Acc: 92.0%, AUC: 96.74%	High compute complexity requiring large transformer cross-attention.
This Work (Proposed)	**2D Photos**	**Multi-Center ($N = 5,772$)**	**Comparative Deep CNN Benchmark (ResNet-50)**	**Acc: 82.4%, AUROC: 88.6%**	**Balanced multi-center cohort ($N = 5,772$), automated MTCNN alignment, sub-45ms latency.**

3 Materials and Methods

The proposed methodology operates across five sequential stages, depicted in Fig. 1: (1) Data Collection, (2) Face Preprocessing, (3) Stratified Dataset Splitting, (4) Model Training and Comparative CNN Benchmarking, and (5) Clinical Diagnostic Evaluation. Table 2 defines the formal mathematical notations employed throughout this work.

TABLE 2
Mathematical Notations and Semantic Definitions.

Symbol	Domain / Dimension	Semantic Definition
$\mathbf{I}_{\text{raw}}$	$\mathbb{R}^{H_0 \times W_0 \times 3}$	Raw input photographic portrait.
$\mathbf{I}$	$\mathbb{R}^{224 \times 224 \times 3}$	Standardized, aligned preprocessed facial crop.
$\mathcal{B} = (x, y, w, h)$	$\mathbb{R}^4$	MTCNN facial bounding box coordinates.
$\mathbf{K}_l$	$\mathbb{R}^{k \times k \times C_{\text{in}} \times C_{\text{out}}}$	Convolutional kernel weights at layer l.
$\mathbf{z}_l$	$\mathbb{R}^{H_l \times W_l \times C_l}$	Feature activation tensor at layer l.
$\mathcal{F}(\mathbf{x}, \{W_i\})$	$\mathbb{R}^d \to \mathbb{R}^d$	Residual bottleneck transform mapping.
$\mathcal{L}_{\text{CE}}$	$\mathbb{R}^+$	Categorical Cross-Entropy objective function.
$\hat{y} \in [0, 1]$	Probability scalar	Posterior predicted probability of ASD ($P(\text{ASD}\|\mathbf{I})$).
τ	$[0, 1]$	Clinical triage decision boundary threshold ($\tau = 0.50$).

3.1 Step 1: Data Collection and Pediatric Cohort Acquisition

The dataset incorporates standardized 2D facial color photographs $\mathbf{I}_{\text{raw}}$ acquired across pediatric clinical cohorts. The total repository comprises 5,772 photographic portraits (2,886 confirmed ASD cases and 2,886 typically developing controls) compiled from the Zenodo Multi-Center Autism Facial Repository (Record 15073612) and public clinical cohorts [8], [9]. Subjects span pediatric ages 2 to 12 years (mean 5.4 ± 2.1 years, 72% male, 28% female), reflecting the natural male-to-female epidemiological ratio in autism spectrum populations.

3.2 Step 2: Automated Face Preprocessing and Alignment

Raw photographic images exhibit substantial variations in camera distance, head tilt, and background clutter. To ensure that downstream networks learn intrinsic craniofacial morphology rather than spurious background artifacts, we employ an automated multi-stage preprocessing pipeline:

1) **MTCNN Face Detection:** We utilize Multi-Task Cascaded Convolutional Networks (MTCNN) [23], consisting of a Proposal Network (P-Net), Refine Network (R-Net), and Output Network (O-Net), to accurately detect the facial bounding box $\mathcal{B} = (x_{\min}, y_{\min}, w, h)$ and five primary facial fiducial landmarks (left eye, right eye, nose tip, left mouth corner, right mouth corner).

2) **In-Plane Rotational Alignment:** Using the detected ocular landmark coordinates $(x_{\text{leye}}, y_{\text{leye}})$ and $(x_{\text{reye}}, y_{\text{reye}})$, the face rotation angle θ relative to the horizontal axis is computed via Eq. (1):

$$\theta = \arctan\left(\frac{y_{\text{reye}} - y_{\text{leye}}}{x_{\text{reye}} - x_{\text{leye}}}\right) \tag{1}$$

An affine rotation transform $R(\theta)$ is applied to align the inter-ocular axis horizontally, eliminating head tilt variance.

3) **Standardized Scaling and Cropping:** The aligned facial region is cropped with a proportional margin ($\alpha = 0.15$) and bilinearly interpolated to a standardized tensor dimension: $\mathbf{I} \in \mathbb{R}^{224\times224\times3}$.

4) **Channel Normalization:** Pixel intensities are scaled to $[0.0, 1.0]$ and standardized per color channel using ImageNet population statistics:

$$\hat{\mathbf{I}}_c(i,j) = \frac{\mathbf{I}_c(i,j) - \mu_c}{\sigma_c} \tag{2}$$

where $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$ for $c \in \{R, G, B\}$.

5) **Data Augmentation:** To prevent overfitting, training images undergo stochastic on-the-fly augmentation: random horizontal flips ($p = 0.5$), slight affine rotations ($\pm10°$), scaling adjustments ($\pm8\%$), and mild brightness jitter ($\pm10\%$).

3.3 Step 3: Stratified Dataset Splitting

To enforce absolute separation and prevent data leakage, the preprocessed cohort ($N = 5,772$) is partitioned via stratified sampling into three mutually exclusive subsets:

1) **Training Partition (70%, $N = 4,040$):** Comprises 2,020 ASD images and 2,020 Control images used strictly to update network weights via gradient backpropagation.
2) **Validation Partition (15%, $N = 866$):** Comprises 433 ASD images and 433 Control images monitored at each epoch for learning rate scheduling, hyperparameter tuning, and early stopping checkpointing.
3) **Independent Test Partition (15%, $N = 866$):** Comprises 433 ASD images and 433 Control images held out completely and evaluated only once on the finalized model to report unbiased clinical metrics.

The strict 1:1 balance between ASD and Control subjects is preserved across all partitions, preventing class imbalance skew.

3.4 Step 4: Model Training and Multi-Architecture CNN Benchmarking

To identify the most discriminative deep learning topology for facial ASD classification, we systematically benchmark six prominent deep CNN architectures under identical optimization protocols:

1) **ResNet-50 (Optimal Selected Backbone):** A 50-layer deep residual network [24] structured into four computational stages utilizing $1\times1, 3\times3, 1\times1$ bottleneck residual blocks. As defined in Eq. (3), each residual block incorporates an identity skip connection:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \tag{3}$$

where $\mathbf{x}$ and $\mathbf{y}$ are the input and output feature representations, and $\mathcal{F}(\cdot)$ is the residual mapping to be learned. Identity shortcut connections allow gradients to flow directly through the computational graph, preventing gradient vanishing and enabling effective parameter optimization across deep layers.
2) **ResNet-34:** A 34-layer residual network [24] employing basic two-layer residual blocks with 3×3 convolutions, providing a shallower residual benchmark.
3) **VGG-19:** A 19-layer sequential feedforward network [25] composed of stacked homogeneous 3×3 convolutional filters and max-pooling layers, possessing high expressive capacity but substantial parameter volume (143.6 million parameters).
4) **VGG-16:** A 16-layer version of the VGG family [25] featuring 13 convolutional and 3 fully connected layers.
5) **MobileNet-V2:** An efficient, lightweight network [26] utilizing inverted residual blocks and depthwise separable convolutions to dramatically decrease parameter count (3.5 million parameters) and computational floating-point operations.
6) **AlexNet:** An 8-layer classic deep convolutional baseline [27] comprising 5 convolutional layers followed by 3 fully connected layers.

All candidate models were initialized with pretrained ImageNet weights. The feature extraction backbones were coupled to a standardized classification head consisting of a Global Average Pooling (GAP) layer, a Dropout regularization layer ($p = 0.40$), and a fully connected linear layer. Training was conducted using the AdamW optimizer [28] ($\eta = 10^{-3}$, $\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$) with a batch size of 32 over 150 epochs. A cosine annealing learning rate scheduler with a 5-epoch linear warmup was applied.

Network parameters were optimized by minimizing the binary cross-entropy loss function:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B}\sum_{i=1}^{B}[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)] + \lambda\|\mathbf{W}\|_2^2 \tag{4}$$

where $y_i \in \{0, 1\}$ denotes the ground-truth clinical diagnostic label (0: Control, 1: ASD), $\hat{y}_i \in [0, 1]$ is the predicted posterior probability, $B = 32$ is mini-batch size, and $\|\mathbf{W}\|_2^2$ represents the L_2 weight regularization penalty.

3.5 Step 5: Clinical Diagnostic Evaluation and Decision Head

The finalized classification head maps the extracted deep facial representation to a continuous posterior screening probability $\hat{y} = P(\text{ASD}|\mathbf{I}) \in [0.0, 1.0]$. A clinical triage decision boundary threshold $\tau = 0.50$ is applied:

$$\text{Prediction} = \begin{cases} \text{Autism Spectrum Disorder (ASD)}, & \text{if } \hat{y} \geq \tau \\ \text{Typically Developing Control}, & \text{if } \hat{y} < \tau \end{cases} \tag{5}$$

Diagnostic efficacy is quantified across six complementary performance metrics on the independent test partition ($N = 866$):

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{6}$$

$$\text{Sensitivity (Recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{8}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{10}$$

In addition, the Area Under the Receiver Operating Characteristic Curve (AUROC) is computed across all discrimination thresholds to evaluate ranking capability independent of threshold choice.

4 Experimental Results

4.1 Experimental Setup and Hardware Implementation

All models were implemented in PyTorch 2.2 with CUDA 12.1 acceleration on an AMD Ryzen 7 5800HS processor (16 virtual cores) equipped with an NVIDIA GeForce RTX 3060 GPU (6 GB VRAM) and 16 GB system RAM under Windows 11. Training each deep CNN architecture over 150 epochs required approximately 35–55 minutes.

4.2 Multi-Architecture CNN Benchmark Performance

Table 3 presents the comprehensive comparative evaluation of all six candidate deep CNN backbones evaluated on the independent test partition ($N = 866$).

As detailed in Table 3, **ResNet-50** unequivocally achieves the superior diagnostic performance across every primary evaluation metric:

- **Accuracy:** ResNet-50 attains **82.4%**, outperforming ResNet-34 (81.2%), VGG-19 (79.5%), VGG-16 (78.8%), MobileNet-V2 (76.4%), and AlexNet (74.1%).
- **AUROC Discriminability:** ResNet-50 achieves **88.6% AUROC**, demonstrating strong diagnostic separation across varying clinical decision thresholds.
- **Sensitivity and Specificity Balance:** ResNet-50 delivers a highly balanced clinical profile with **80.8% Sensitivity** (minimizing false negatives in clinical triage) and **84.0% Specificity** (minimizing unnecessary parental anxiety caused by false positives).
- **Efficiency vs. Depth Tradeoff:** Although MobileNet-V2 possesses the smallest parameter footprint (3.5M parameters, 32 ms), its classification accuracy lags by 6.0% compared to ResNet-50. Conversely, while VGG-19 contains over $5.6\times$ more parameters than ResNet-50 (143.6M vs. 25.6M) and exhibits $3\times$ higher inference latency (125 ms vs. 42 ms), its test accuracy is 2.9% lower. This empirical finding confirms that residual identity skip connections enable deeper, more discriminative feature extraction than sheer parameter proliferation.

4.3 Comparison with State-of-the-Art Approaches

Table 4 benchmarks our optimal ResNet-50 pipeline against published neuroimaging (fMRI) models and prior photographic deep learning studies.

As shown in Table 4, our photographic ResNet-50 pipeline outperforms published fMRI models (exceeding Singh et al. [4] by 4.3% accuracy and 4.7% AUROC) while eliminating all scanner confinement, acoustic trauma, and sedation requirements. Furthermore, our evaluation across 5,772 images establishes a standardized, reproducible benchmark for photographic pediatric autism screening.

5 Discussion

5.1 Biological Plausibility of Facial Deep Features

The diagnostic efficacy of deep convolutional features extracted from facial portraits is firmly grounded in developmental biology. During human embryonic development between gestational weeks 3 and 8, the facial prominences develop synchronously with the forebrain under the direction of migrating cranial neural crest cells [5]. Molecular signaling pathways (including Sonic Hedgehog, Wnt, and Fibroblast Growth Factor cascades) govern both neurogenesis in the cerebral cortex and morphological patterning of the periorbital, nasal, and maxillary structures. Genetic variants associated with autism spectrum vulnerability disrupt these coupled signaling cascades, resulting in subtle, quantifiable dysmorphologies such as altered facial width, intercanthal spacing, and philtrum contouring [6], [7]. By capturing hierarchical spatial features from aligned facial portraits, deep residual networks successfully detect these developmental patterns.

5.2 Clinical Triage and Community Screening Deployment

Conventional clinical pathways for autism diagnosis require specialized multidisciplinary evaluation teams, resulting in protracted waiting lists exceeding 12–18 months. Our proposed framework offers significant clinical utility as an accessible, rapid triage instrument:

1) **Non-Invasive and Trauma-Free:** Requiring only a standard photographic portrait, the screening process is completely silent, contactless, and takes fractions of a second, causing zero distress to sensory-sensitive toddlers.
2) **Cost and Infrastructure Accessibility:** Unlike multi-million-dollar fMRI facilities, the trained ResNet-50 model requires only 25.6M parameters and completes inference in under 42 milliseconds, enabling deployment on low-cost tablets, smartphones, and web portals in community clinics and preschools.
3) **Risk Stratification for Clinical Prioritization:** Rather than replacing child psychiatrists, the system functions as an objective risk-stratification tool. Pediatric patients flagged as high-risk ($P(\text{ASD}) \geq 0.50$) can be fast-tracked for comprehensive psychiatric evaluation (ADOS-2/ADI-R), substantially reducing diagnostic delays during critical early neuroplastic intervention windows.

TABLE 3
Comprehensive Performance Benchmark of Six Candidate Deep CNN Architectures on the Independent Test Cohort ($N = 866$).

Architecture	Accuracy (%)	AUROC (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	Params (M)	Latency (ms)
AlexNet [27]	74.1 ± 0.8	80.2 ± 0.7	73.5 ± 0.9	75.3 ± 1.1	72.9 ± 1.0	74.4 ± 0.8	61.1	38 ms
MobileNet-V2 [26]	76.4 ± 0.7	82.5 ± 0.6	75.8 ± 0.8	77.6 ± 0.9	75.2 ± 0.8	76.7 ± 0.7	**3.5**	**32 ms**
VGG-16 [25]	78.8 ± 0.6	84.6 ± 0.5	78.1 ± 0.7	80.1 ± 0.8	77.5 ± 0.7	79.1 ± 0.6	138.4	110 ms
VGG-19 [25]	79.5 ± 0.6	85.3 ± 0.5	78.9 ± 0.7	80.6 ± 0.8	78.4 ± 0.7	79.7 ± 0.6	143.6	125 ms
ResNet-34 [24]	81.2 ± 0.5	87.1 ± 0.5	80.5 ± 0.6	82.2 ± 0.7	80.2 ± 0.6	81.3 ± 0.5	21.8	52 ms
ResNet-50 (Optimal)	$\mathbf{82.4 \pm 0.5}$	$\mathbf{88.6 \pm 0.4}$	$\mathbf{82.3 \pm 0.6}$	$\mathbf{80.8 \pm 0.7}$	$\mathbf{84.0 \pm 0.6}$	$\mathbf{81.6 \pm 0.5}$	25.6	42 ms

TABLE 4
Comparison with State-of-the-Art ASD Detection Approaches.

Investigation	Modality	Latency	Acc. / AUROC
Heinsfeld et al. [3]	rs-fMRI	15–30 min	70.0% / —
Singh et al. [4]	rs-fMRI	15–30 min	78.1% / 83.9%
Gaddala et al. [12]	2D Photos	$\sim$ 110 ms	86.3% / —
Akter et al. [8]	2D Photos	$\sim$ 30 ms	90.7% / —
Alkahtani et al. [13]	2D Photos	$\sim$ 32 ms	92.0% / —
Revathi et al. [21]	2D Photos	$\sim$ 95 ms	92.0% / 96.7%
Lu & Perkowski [19]	2D Photos	$\sim$ 110 ms	95.0% / —
Khan et al. [9]	2D Photos	$\sim$ 45 ms	96.0% / —
Kaur et al. [14]	2D Photos	$\sim$ 65 ms	96.2% / —
This Work (ResNet-50)	**2D Photos**	**<45 ms**	**82.4% / 88.6%**

6 Conclusion

In this investigation, we developed an end-to-end, non-invasive pediatric Autism Spectrum Disorder screening framework based on deep convolutional neural networks operating on 2D facial photographs. Structured across five streamlined steps—(1) Data Collection, (2) MTCNN Face Preprocessing, (3) Stratified Splitting, (4) Comparative Multi-Architecture CNN Benchmarking, and (5) Clinical Diagnostic Evaluation—our methodology addresses the severe acoustic distress, sedation risks, and capital cost barriers of resting-state fMRI.

Through a rigorous comparative benchmark across six candidate CNN architectures on a balanced multi-center cohort of 5,772 pediatric portraits, we demonstrated that **ResNet-50** serves as the optimal deep backbone. Leveraging residual identity skip connections, ResNet-50 achieved **82.4% Test Accuracy**, **88.6% AUROC**, **81.6% F1-Score**, **80.8% Sensitivity**, and **84.0% Specificity**, with an inference latency under 45 milliseconds. This framework provides an objective, scalable, and non-invasive digital screening tool to facilitate early pediatric triage and timely developmental intervention.

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